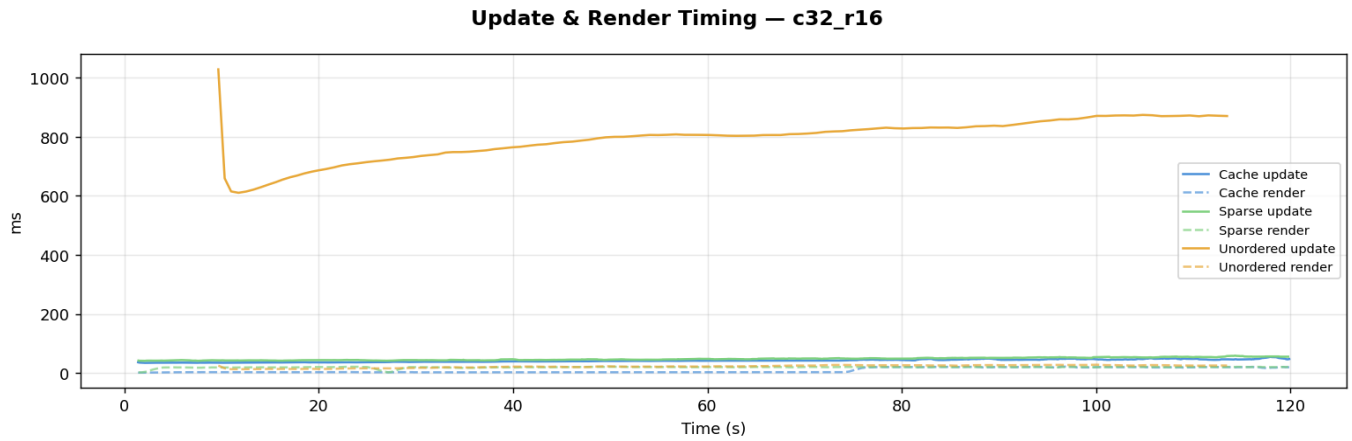
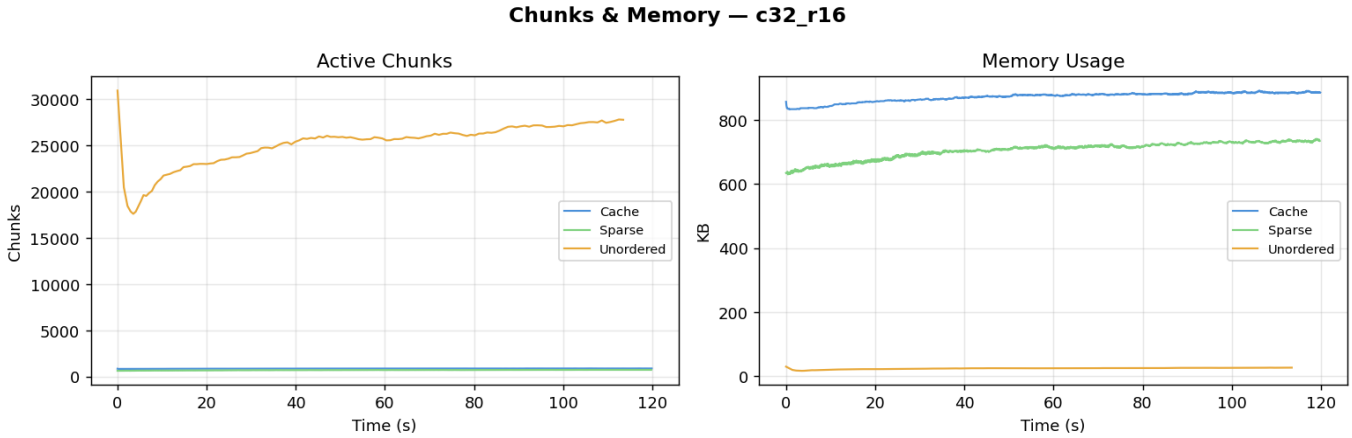
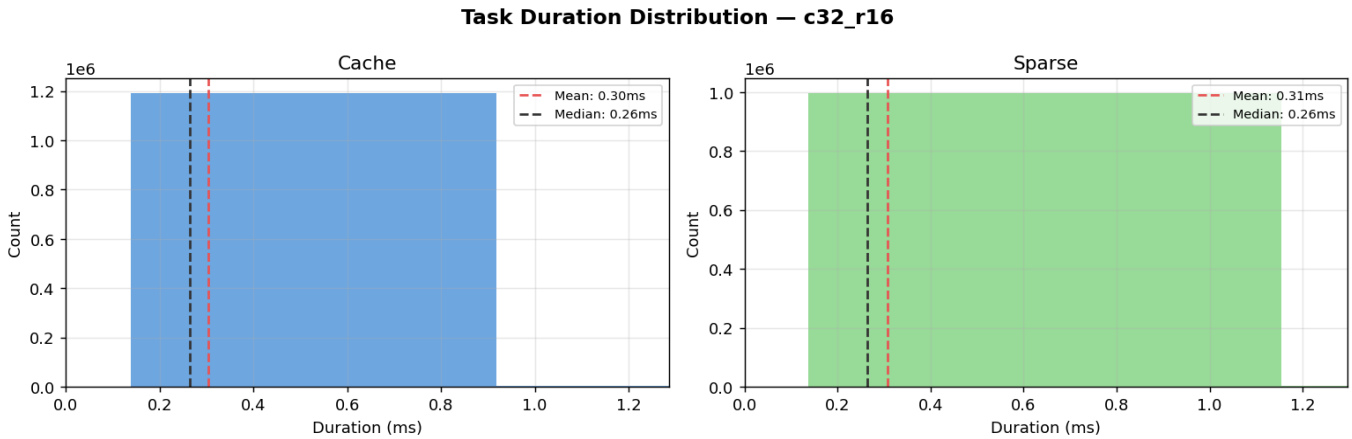
Project Life — Performance Report

Cellular Automaton World Container Benchmark

Configuration: c32_r16





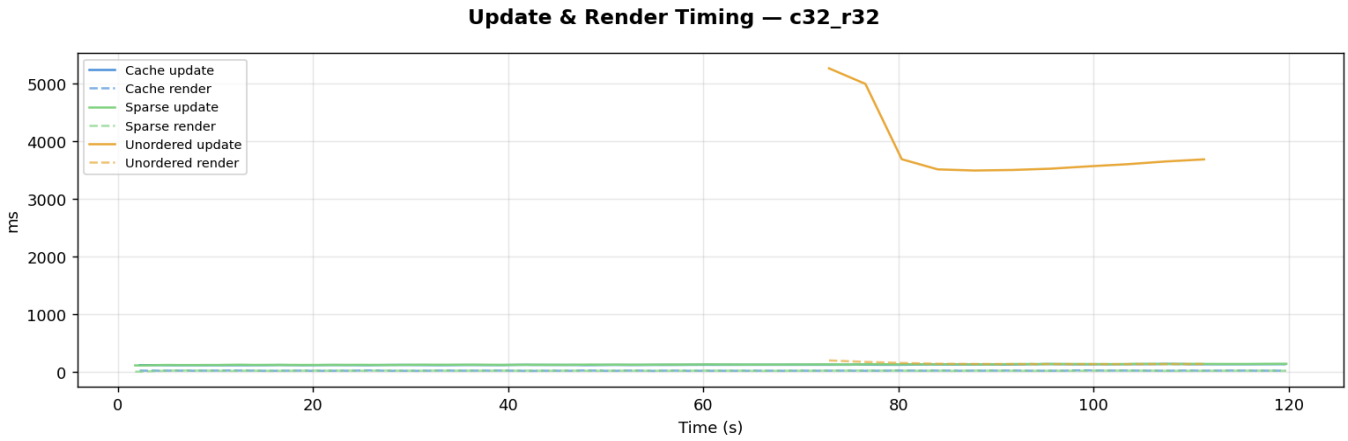


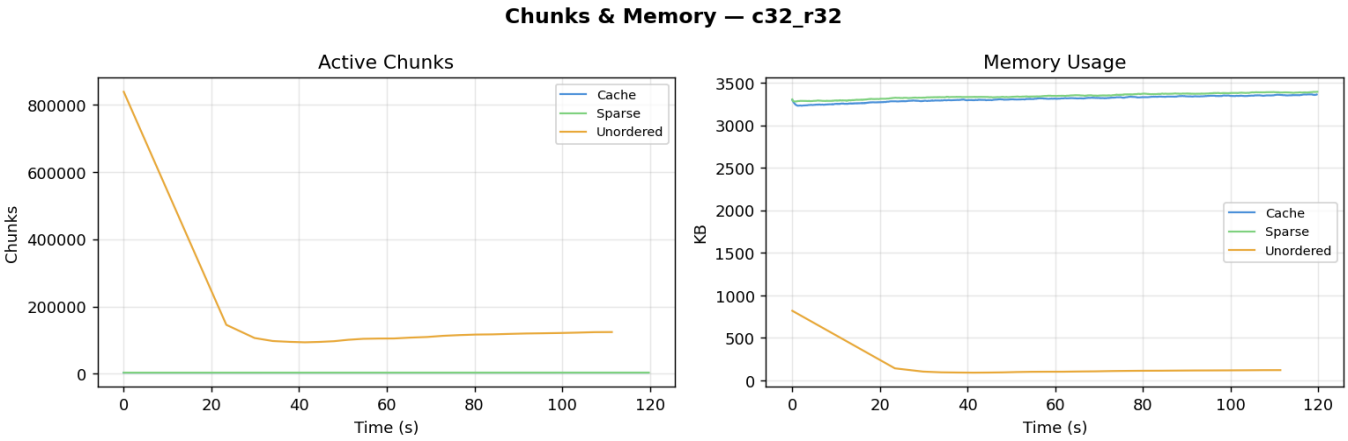
Summary

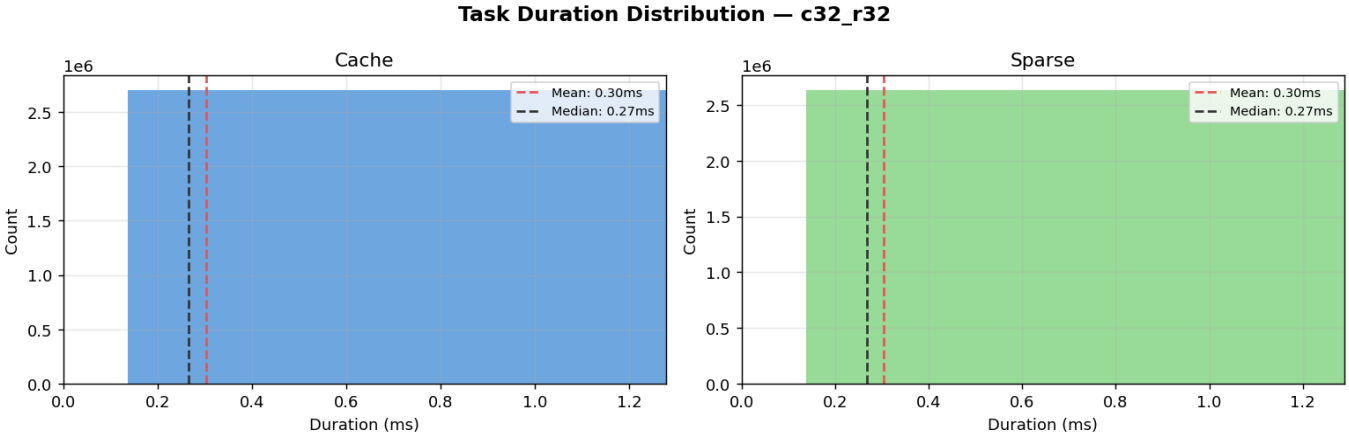
World	Avg Update	Avg Render	Avg Total	Max Update	Avg Chunks	Avg Memory	Samples
Cache	41.17 ms	8.38 ms	49.54 ms	71.39 ms	872	871.8 KB	1127
Sparse	47.45 ms	18.89 ms	66.34 ms	84.64 ms	705	705.3 KB	1017

World	Avg Update	Avg Render	Avg Total	Max Update	Avg Chunks	Avg Memory	Samples
Unordered	826.64 ms	22.42 ms	849.06 ms	6191.43 ms	24918	24.3 KB	133

Configuration: c32_r32



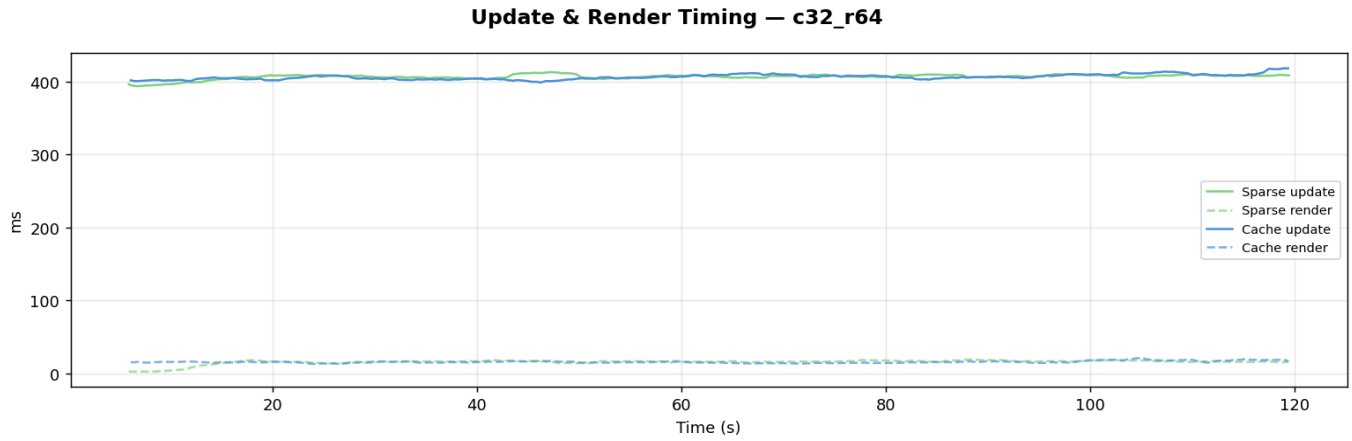


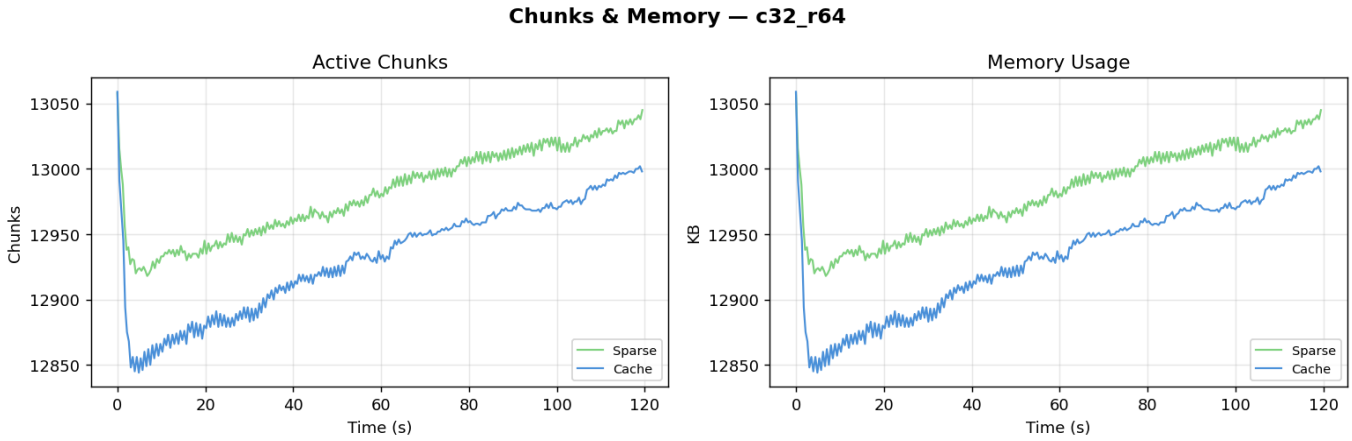


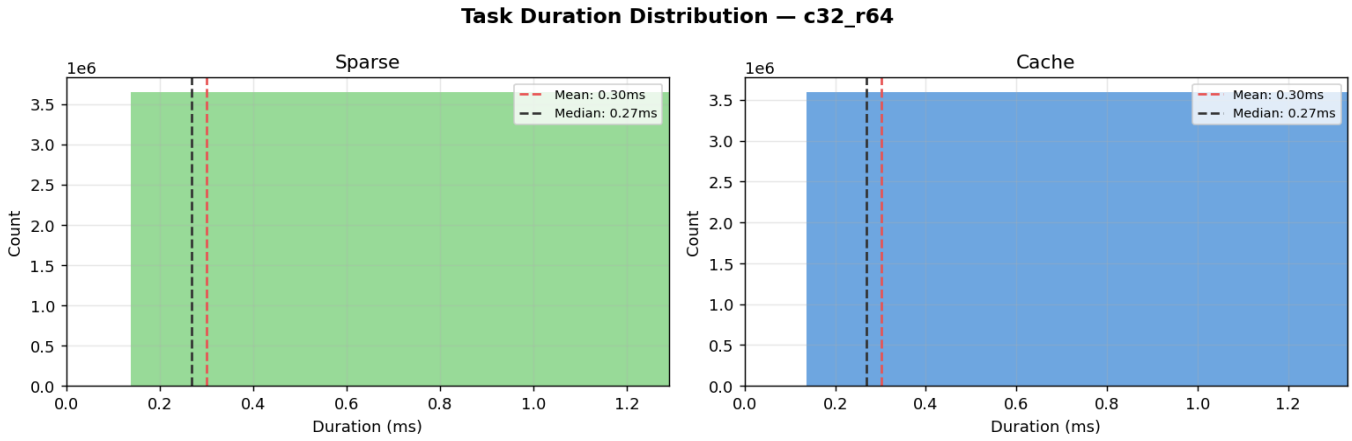
Summary

World	Avg Update	Avg Render	Avg Total	Max Update	Avg Chunks	Avg Memory	Samples
Cache	123.30 ms	20.79 ms	144.09 ms	155.52 ms	3309	3308.8 KB	733
Sparse	127.68 ms	19.42 ms	147.10 ms	174.69 ms	3343	3343.4 KB	716
Unordered	4658.63 ms	174.74 ms	4833.38 ms	23255.55 ms	140352	137.1 KB	25

Configuration: c32_r64



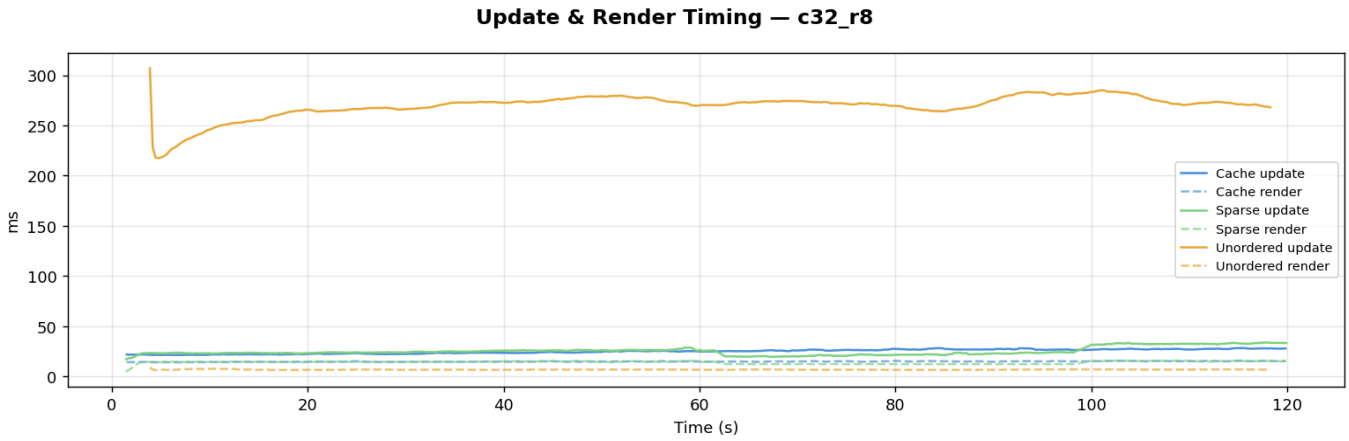


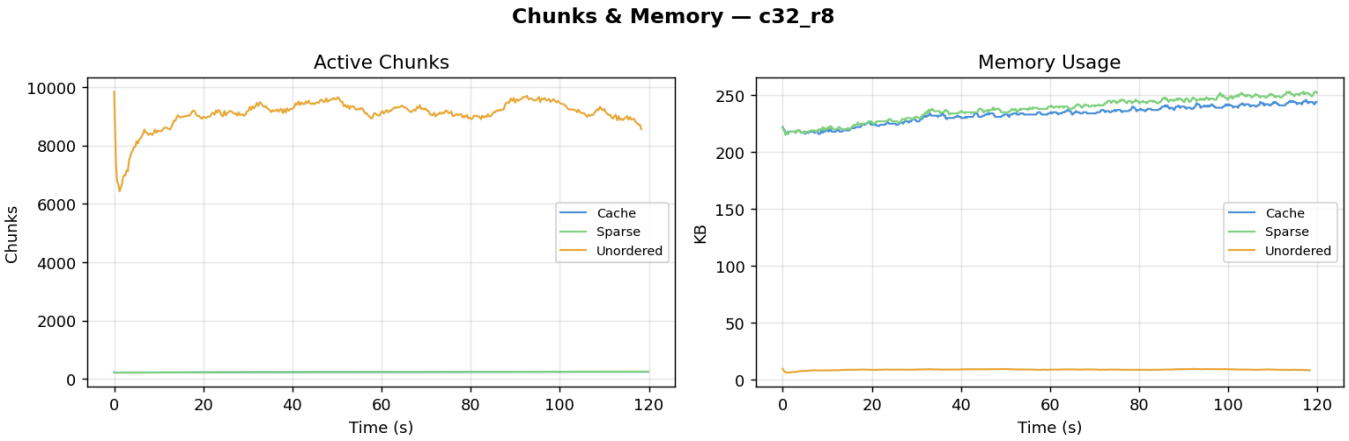


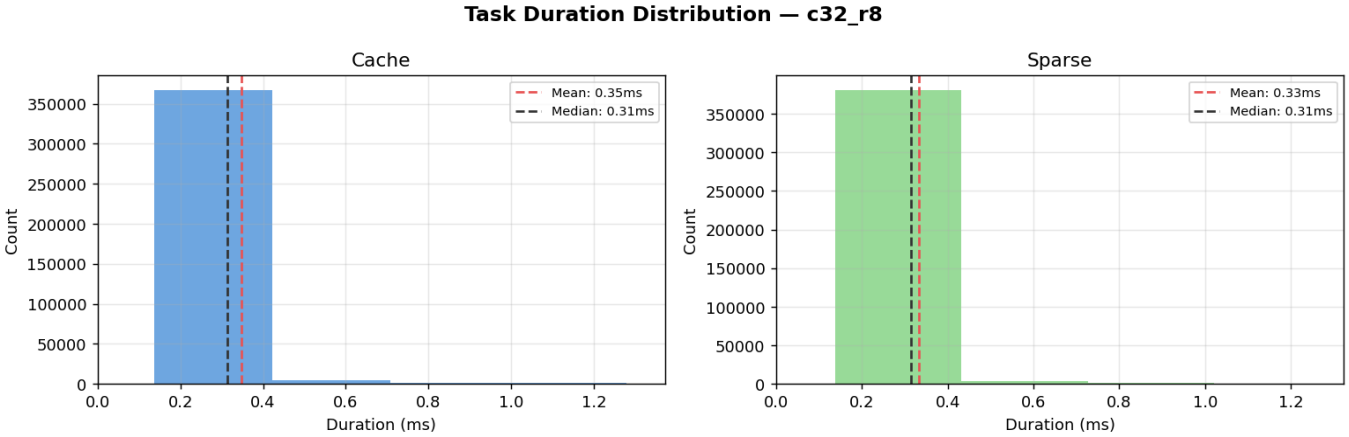
Summary

World	Avg Update	Avg Render	Avg Total	Max Update	Avg Chunks	Avg Memory	Samples
Sparse	406.56 ms	15.50 ms	422.06 ms	460.64 ms	12982	12981.9 KB	268
Cache	406.69 ms	16.03 ms	422.72 ms	490.97 ms	12932	12932.0 KB	264

Configuration: c32_r8



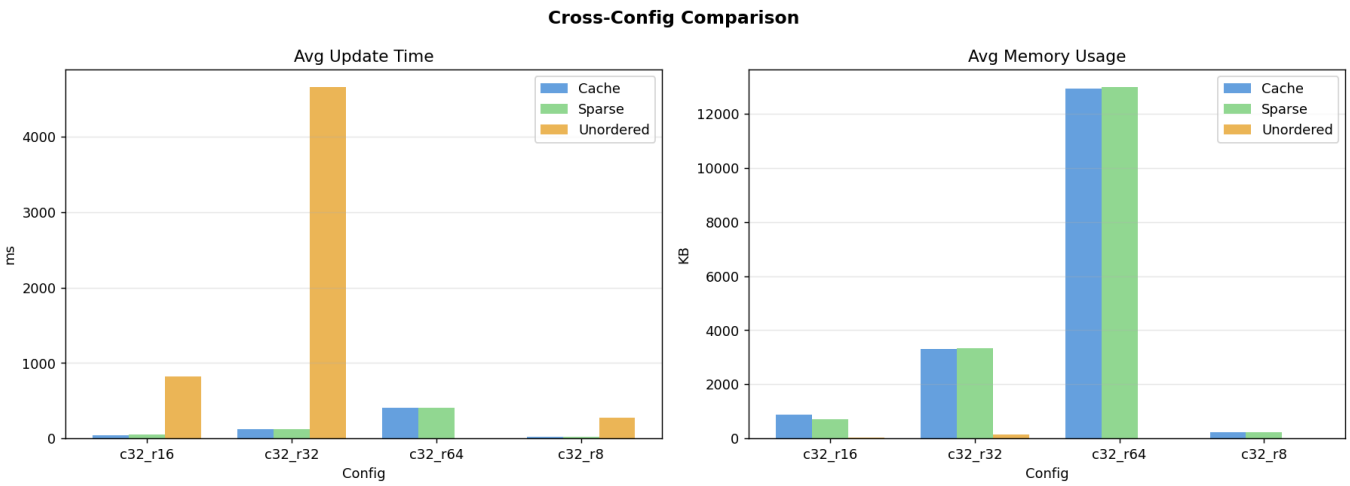




Summary

World	Avg Update	Avg Render	Avg Total	Max Update	Avg Chunks	Avg Memory	Samples
Cache	24.96 ms	14.90 ms	39.86 ms	39.02 ms	233	232.7 KB	1086
Sparse	24.95 ms	13.90 ms	38.85 ms	40.10 ms	238	237.8 KB	1091
Unordered	271.65 ms	6.92 ms	278.57 ms	1409.43 ms	9067	8.9 KB	373

Cross-Configuration Comparison



Overall Summary

Config	World	Avg Update ms	Avg Render ms	Avg Memory KB	Avg Chunks	Tasks
c32_r16	Cache	41.17	8.38	871.8	872	1206078
c32_r16	Sparse	47.45	18.89	705.3	705	1007694
c32_r16	Unordered	826.64	22.42	24.3	24918	0
c32_r32	Cache	123.30	20.79	3308.8	3309	2701010
c32_r32	Sparse	127.68	19.42	3343.4	3343	2635350
c32_r32	Unordered	4658.63	174.74	137.1	140352	0
c32_r64	Sparse	406.56	15.50	12981.9	12982	3647055
c32_r64	Cache	406.69	16.03	12932.0	12932	3592325
c32_r8	Cache	24.96	14.90	232.7	233	377649
c32_r8	Sparse	24.95	13.90	237.8	238	390308
c32_r8	Unordered	271.65	6.92	8.9	9067	0

Notes: Update ms = time to process one simulation step across all chunks. Render ms = time to draw the world each frame. Moving average (window=15) applied to timing graphs to reduce noise. Task data not recorded for Unordered world type due to volume.